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融合思維運用於增強大型語言模型之結構化推理能力

Fusion of Thoughts for Enhancing Structured Reasoning
in Large Language Models

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Large Language Models

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論文提要內容：

由於結構化推理路徑能顯著提升大型語言模型 (LLMs) 的推理能力，近年來，思維鏈 (Chain of Thought, CoT)、思維樹 (Tree of Thought, ToT) 和思維圖 (Graph of Thought, GoT) 等提示策略受到極大關注。然而，現有方法常將每個中間思維視為獨立單元，忽略了整合多個思維以產生更連貫、資訊更豐富的提示的潛力。本論文中，我們針對思維在推理過程中的語義融合，開發了一種新穎的提示框架：思維融合 (Fusion of Thought, FoT)。該框架引入了動態融合機制，不僅考慮了現有的多個獨立思維，還將這些思維融成品質更高的「融合思維」，並將其作為後續推理步驟的新輸入。為了最大化推理鏈的深度和廣度並提高準確性，我們基於思維融合建立了一個最佳化問題。此最佳化問題包括思維選擇、思維融合與推理路徑設計。實驗結果表明，我們提出的 FoT 方案相比現有方法能有效提高答案準確性並減少冗餘推理路徑。

關鍵字：思維鏈、思維樹、思維圖、大型語言模型、提示工程。

ABSTRACT

Fusion of Thoughts for Enhancing Structured Reasoning in Large Language Models

by

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July, 2025

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DEPARTMENT: COMPUTER SCIENCE & INFORMATION ENGINEERING

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In recent years, the development of prompting strategies such as Chain-of-Thought (CoT), Tree-of-Thought (ToT), and Graph-of-Thought (GoT) have demonstrated that structured reasoning paths can significantly enhance the reasoning capabilities of large language models (LLMs). However, existing methods often treat each intermediate thought as an independent unit, overlooking the potential of integrating multiple thoughts to produce more coherent and informative prompts. This thesis proposes a novel prompting framework, Fusion of Thought (FoT), which introduces a dynamic fusion mechanism to semantically combine related thoughts during the reasoning process. The resulting "fused thoughts" are of higher quality and are used as new inputs for subsequent reasoning steps. This mechanism not only expands the depth and breadth of the reasoning chain but also integrates naturally with a variety of existing prompting architectures. Experiments conducted on both synthetic and open-domain reasoning tasks show that FoT outperforms existing prompting methods in terms of answer accuracy and reduction of redundant reasoning paths. These results confirm the effectiveness of thought fusion as a key strategy for improving prompt design in complex reasoning tasks.

Key words: Chain of Thought, Tree of Thought, Graph of Thought, Large Language Models, Prompt Engineering.

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Chapter 1

Introduction

As large language models (LLMs) such as GPT-4 continue to develop, they have demonstrated impressive capabilities in natural language understanding and generation. Bubeck *et al.* [1] explored the signs of artificial general intelligence emerging from early GPT-4 experiments, which showed the potential for reasoning, abstraction, and knowledge transfer. OpenAI's official report [2] further confirmed the robustness of GPT-4 through extensive evaluation of language, mathematics, and coding capabilities. Oswald Campesato [3] emphasized the importance of theoretically integrating symbolic logic into LLMs to enhance their combinatorial reasoning capabilities, laying the foundation for building cognitive style models. Prompt engineering has become a key area for improving the multi-step reasoning capabilities of LLMs. Pranab Sahoo *et al.* [4] proposed a taxonomy of prompt engineering techniques, covering categories such as zero-shot, instruction tuning, and chain of thought (CoT). Sander Schulhoff *et al.* [5] established a comprehensive "Prompt Report" to evaluate aspects such as prompt reuse, prompt sharing, and interpretability. In addition, Kaito Yamamoto *et al.* [6] introduced a tool integration prompt strategy that aims to modularize the reasoning process of LLMs for external tools and functions, which further expands the capabilities of LLMs in practical applications. As the size of language models increases, their reasoning performance has been significantly improved by introducing structured hint frameworks, such as Chain-of-Thought (CoT), Tree-of-Thought (ToT), and Graph-of-Thought (GoT). These methods allow LLMs to decompose complex problems into intermediate reasoning steps, thereby improving

interpretability and problem-solving efficiency. CoT significantly improves the performance of complex reasoning tasks by guiding the model to generate a series of intermediate thoughts to solve the problem. ToT further organizes the reasoning process into a tree structure, allowing the model to explore and backtrack in multiple potential thought paths, thereby improving the robustness of the solution. GoT organizes thoughts into a more flexible graph structure that can capture complex relationships and nonlinear dependencies between thoughts, providing solutions for more complex problems. However, despite the effectiveness of these methods, existing methods often treat each intermediate thought as an isolated unit, lacking a mechanism to unify semantically related thoughts, which in turn limits the coherence of reasoning, which may lead to information redundancy or fail to fully utilize the potential synergy between different thoughts. However, existing methods tend to treat each intermediate thought as an independent unit, with limited exploration of how multiple thoughts can be integrated to form a more coherent and information-rich basis for reasoning. This limitation may result in redundant, repetitive, or even contradictory reasoning paths, ultimately affecting the quality of the final output.

To address this issue, this thesis proposes a novel prompting framework, Fusion of Thought (FoT). As illustrated in figure 1.1, the core idea of FoT is to dynamically fuse multiple semantically related candidate thoughts during the reasoning process to generate high-quality fused thoughts, which are then used as prompt inputs for subsequent reasoning. This mechanism enhances semantic integration and improves the logical coherence of the reasoning trajectory.

The contributions of this thesis is organized as follows.

- To enhance LLM reasoning, we propose Fusion of Thought (FoT). This innovative prompting strategy dynamically merges semantically similar intermediate thoughts, regardless of their initial quality, to generate more robust and coherent inputs during the reasoning process. This fusion produces more logically coherent and contextually relevant thoughts, thereby enhancing the overall quality and stability of the reasoning process.

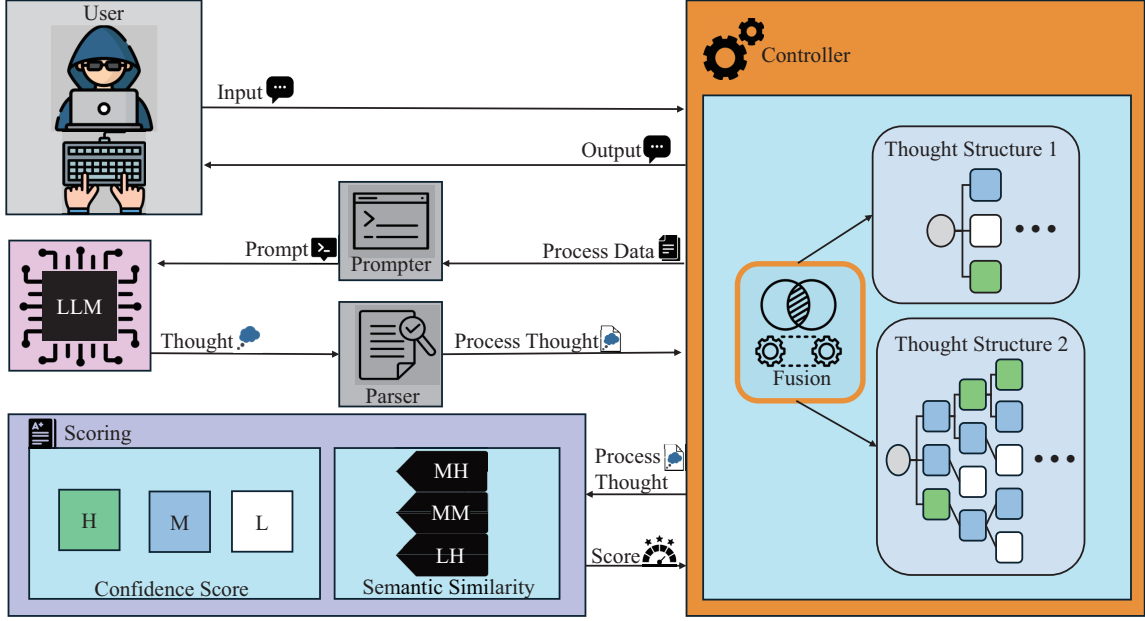


Figure 1.1: Our proposed Fusion of Thought scheme.

- FoT can be seamlessly integrated into various reasoning frameworks, such as Tree-of-Thought and Graph-of-Thought. While preserving the original structure’s exploratory capacity, it introduces a flexible fusion mechanism that enables complementary recombination of thoughts with different confidence levels. This promotes information integration and semantic convergence, improving the logical consistency and potential of the reasoning process.
- Experiments conducted on both synthetic and open-domain reasoning tasks demonstrate that FoT outperforms existing prompting methods in terms of reasoning accuracy and thought expansion efficiency, validating thought fusion as an effective approach for enhancing the reasoning performance of large language models.

The rest of the thesis is organized as follows. chapter 2 describes related work and motivation. chapter 3 presents the system model, problem formulation, and basic concepts. chapter 4 tells Fusion of Thought (FoT) framework. chapter 5 presents experimental results and analysis. Finally, chapter 6 concludes the thesis with a summary of the work and a discussion of future research directions.

Chapter 2

Research Background

This chapter describes the related works in section 2.1 and discusses the motivation for our research in section 2.2.

2.1 Related works

In recent years, prompt engineering has played a pivotal role in guiding large language models (LLMs) to produce structured reasoning behaviors. Pranab Sahoo *et al.* [4] and Sander Schulhoff *et al.* [5] offered comprehensive surveys on emerging prompting techniques and frameworks, not only proposing detailed taxonomies but also establishing practical metrics for evaluating their impact.

Kaito Yamamoto *et al.* [6] introduce a framework that leverages evolutionary strategies to automatically refine natural language prompts for large language models. In this approach, prompts are treated as individuals in an evolutionary process, undergoing selection, mutation, and fitness evaluation to gradually evolve into more effective forms. The optimized prompts are shown to guide models toward improved reasoning behaviors, such as decomposing complex problems, performing step-by-step inference, and even leveraging external tools when necessary. Experimental results demonstrate significant performance gains across diverse reasoning tasks, highlighting the potential of prompt evolution as a lightweight yet powerful alternative to retraining.

Pei Zhou *et al.* [7] proposed the self-discover framework, which enables LLMs to au-

tonomously compose reasoning structures by selecting and assembling modular reasoning components. Unlike fusion-based approaches such as our FoT, which focus on semantically integrating multiple intermediate thoughts, self-discover emphasizes structural induction by allowing the model to decide *how* reasoning steps should be organized. This complementary perspective highlights two parallel directions: fusion strategies that consolidate diverse thoughts into coherent solutions, and structure-induction methods that dynamically compose task-specific reasoning pathways.

In the domain of structured reasoning, a landmark advancement was introduced by Xuezhi Wang *et al.* [8] through the Chain-of-Thought (CoT) prompting method, which enables LLMs to simulate the step-by-step reasoning process akin to human cognition. By instructing the model to generate intermediate steps, CoT significantly improves performance in solving complex problems. Xuezhi Wang *et al.* [8] enhanced this approach by introducing Self-Consistency, which reduces output variability through multiple sampling and aggregation. Zayne Sprague *et al.* [9] further analyzed CoT’s applicability and concluded that it is particularly effective in symbolic and arithmetic reasoning tasks. Jianwei Yang *et al.* [10] proposed Faithful-CoT, integrating uncertainty estimation to mitigate hallucinations and improve the trustworthiness of intermediate reasoning steps. To further enhance control and interpretability, Fan Wang *et al.* [11] introduced Contrastive Prompt Tuning for controllable CoT behavior. To expand reasoning flexibility and exploratory depth, Shunyu Yao *et al.* [12] proposed the Tree-of-Thought (ToT) framework, which organizes reasoning as a tree-based search process, enabling backtracking and evaluation over multiple paths. Jieyi Long [13] extended this framework by allowing LLMs to autonomously guide the expansion of reasoning branches, increasing both breadth and depth. Building upon this, Besta *et al.* [14] generalized the structure to a Graph-of-Thought (GoT) model, facilitating more complex multi-path traversals and non-linear inference. Bowen Jin *et al.* [15] enhanced GoT by integrating CoT-style reasoning into graph-based structures with richer semantic context, thus improving its capacity to tackle intricate problems. Haotian Zhao *et al.* [16] explored early designs of Graph-of-Thoughts that

focus on thought aggregation and pathway selection in non-linear reasoning.

Beyond structural prompting, researchers have actively explored fusion-based strategies to improve reasoning quality. Xueyang Xing *et al.* [17] proposed a method combining knowledge graphs with causal injection mechanisms, significantly improving the model’s ability to ground its reasoning in factual knowledge. Fanqi Wan *et al.* [18] introduced a multi-source knowledge fusion method that aggregates complementary reasoning from diverse perspectives and sources, resulting in more comprehensive and accurate inferences. Prompt Voting, introduced by Kevin Lee *et al.* [19], decodes robustness of LLM reasoning via ensemble voting, showing improvements in consistency under adversarial prompts. These foundational works collectively inform the motivation behind our Fusion-of-Thought (FoT) framework, which aims to semantically integrate intermediate thoughts to enhance logical coherence and reasoning depth.

Recent studies have also investigated confidence estimation and multi-agent architectures as means to regulate and verify reasoning outputs. Yuxi Xie *et al.* [20] introduced a self-evaluation guided beam search that allows models to self-calibrate and refine output candidates during generation. Tian Xia *et al.* [21] incorporated cross-model consensus mechanisms to eliminate inconsistent outputs and improve result reliability. Aohan Zeng *et al.* [22] proposed Debate-CoT, an innovative multi-agent system that simulates inter-model debates to explore diverse reasoning paths and select optimal conclusions. These contributions collectively underscore the significance of fusion, feedback, and adaptive reasoning in the development of robust LLM capabilities for complex decision-making and problem solving. Yujia Zheng *et al.* [23] introduced the LLM-as-a-Judge paradigm, utilizing pairwise ranking to autonomously evaluate and filter generated responses.

In summary, the existing literature has laid a strong foundation for structured and modular reasoning in LLMs, and recent progress has enabled the exploration of diverse reasoning paths and result validation. However, current methods still lack a dynamic and semantically aware fusion mechanism capable of integrating thoughts from different quality levels and sources

while maintaining logical coherence throughout the reasoning process. Our proposed FoT framework positions itself as a comprehensive strategy that unifies diverse reasoning trajectories. By actively fusing semantically related thoughts, it complements and extends existing paradigms—ultimately achieving more cohesive, efficient, and accurate LLM reasoning.

2.2 Motivation

While existing prompting strategies such as Chain-of-Thought (CoT) [7], Tree-of-Thought (ToT) [11], and Graph-of-Thought (GoT) [13] have significantly enhanced the structured reasoning abilities of large language models (LLMs), they each exhibit critical limitations (as shown in figure 2.1). CoT is effective for linear symbolic reasoning but struggles with ambiguity and contradictions, even when enhanced by self-consistency sampling to reduce variability. ToT expands reasoning coverage through multi-branch exploration, yet it treats each intermediate thought as an isolated unit and relies heavily on score-based pruning, often resulting in the premature elimination of potentially valuable low or mid quality thoughts. GoT generalizes the structure into a graph, offering more flexible multi-path traversal, but it lacks controllable fusion strategies and does not differentiate the quality of individual thoughts, which can lead to semantic drift, redundancy, and inflated inference costs.

Beyond structural prompting, recent work such as self-discover [23] has introduced the idea of structural induction, enabling LLMs to autonomously compose reasoning structures by selecting and assembling modular components. This approach provides greater flexibility in determining how reasoning steps should be organized, but it still emphasizes structural composition rather than semantic integration. Similarly, efforts such as evolutionary prompt optimization [6] and fusion-based reasoning methods [16,17] have shown promise in improving adaptability and factual grounding. However, these strategies either optimize prompts at the surface level or fuse outputs through heuristic aggregation, falling short of delivering a dynamic, semantically aware mechanism that can coherently consolidate intermediate reasoning across varying quality levels.

In practice, this gap manifests as a tension between diversity and coherence: methods that encourage structural exploration often achieve broader coverage but suffer from fragmented or inconsistent reasoning, while methods that aggregate results risk oversimplification and loss of nuanced intermediate insights. Moreover, existing approaches rarely differentiate the confidence and reliability of thoughts, treating all candidates as equally valuable, which can propagate noise and contradictions into the final output.

Therefore, the core motivation of this thesis is to propose a new prompting framework Fusion-of-Thought (FoT)—that flexibly combines the advantages of structured reasoning while introducing a semantic fusion mechanism. By selectively integrating semantically consistent and complementary thoughts based on confidence and similarity, FoT significantly improves cohesion, logical consistency, and efficiency of reasoning, successfully bridging the gap between reasoning diversity and semantic convergence.

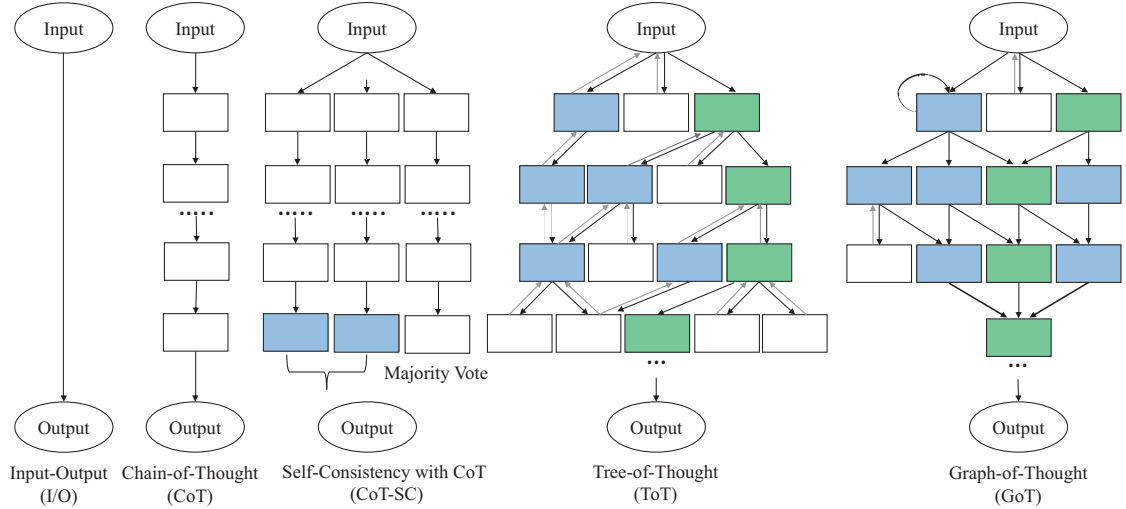


Figure 2.1: Comparison of reasoning strategies: CoT uses single-path reasoning, SC-CoT aggregates multiple reasoning paths, ToT generates reasoning paths based on a tree structure, and GoT implements graph-based traversal.

Chapter 3

Preliminaries

This chapter describes the system model, define the problem formulation, and explain the basic idea in sections 3.1, 3.2, and 3.3, respectively.

3.1 System Model

The Fusion of Thoughts (FoT) architecture consists of a set of interacting modules that work together to support multi-step reasoning and thought fusion. As shown in figure 1.1, the system consists of four core components: prompter, parser, scorer, and controller. Together, these modules transform the task input x into a reasoning process with logical coherence and semantic structure. The modular design of FoT provides high flexibility, enabling it to be applied to a variety of reasoning tasks and support different structural forms, such as tree-based or graph-based representations.

The prompter module is responsible for transforming the raw input x into prompt samples suitable for querying the Large Language Model (LLM). These prompts are dynamically constructed according to the task context and are used to drive the LLM to generate multiple sets of candidate thoughts, promote the exploration of different reasoning paths, and thus guide the model to produce different intermediate results. The parser module is responsible for converting the textual responses output by the LLM into structured thought units. These thought units are standardized and indexed for subsequent evaluation, tracking, and multi-round use. This structured representation is essential for managing the branching, in-

tegration, and evolution of thoughts. The scoring module performs two main tasks. First, for each thought unit, it computes an online proxy confidence score without relying on gold answers. This score integrates multiple signals, including peer self-consistency, retrieval overlap, lightweight entailment checks, and penalties for uncertainty or formatting. Based on the score, each thought is classified into three confidence bands: low (L), medium (M), and high (H). In practice, the low-confidence band covers a wide range, where extremely low-scoring thoughts are directly preserved without attempting fusion, while the remaining low-confidence thoughts may participate in fusion operations. Second, this module calculates the semantic similarity between thought units using embedding-based similarity or lexical overlap. This similarity information, together with the confidence bands, provides the basis for subsequent thought fusion operations. The controller module coordinates the entire reasoning process. It dynamically maintains the reasoning structure (such as a search tree or reasoning graph) and determines how to integrate newly generated ideas into the existing structure. Based on the confidence bands and similarity indicators provided by the scoring module, the controller selects candidate thought combinations and applies one of six fusion strategies: high-high (HH), medium-medium (MM), low-low (LL), medium-high (MH), low-high (LH), or low-medium (LM). These fused thoughts are then evaluated together with the original thoughts and integrated into the reasoning structure when appropriate. Through this mechanism, FoT gradually enhances the semantic coherence and credibility of the reasoning trajectory, while pruning redundant or low-quality branches. The interaction between these modules constitutes the core logic of the FoT framework, making it highly flexible while maintaining a rigorous structure. By clearly distinguishing between operations such as generation, parsing, evaluation, fusion, and control, FoT can be widely adapted to different task scenarios and integrated with a variety of language models and prompting strategies.

3.2 Problem formulation

In this section, FoT let x denote the input of the task, which can be any semantically structured form, such as prompts, statements, or questions. The objective of the Fusion of Thoughts (FoT) framework is to derive the most representative final thought from a set of intermediate thoughts through multiple rounds of thought generation, evaluation, and semantic fusion, ultimately yielding a coherent task solution. The reasoning process is supported by a large language model p_θ . At each reasoning step $t \in \{1, 2, \dots, T\}$, the thought generator $G(p_\theta, \cdot)$ expands every thought node $s \in S_{t-1}$ into at most k new candidate thoughts. The overall thought set produced at round t is therefore given by

$$G_t = \bigcup_{s \in S_{t-1}} G(p_\theta, s, k). \quad (3.1)$$

Once the candidates are generated, each thought $t \in G_t$ is evaluated by the scoring function $V(\cdot)$, which assigns a normalized confidence score $\text{conf}(t) \in [0, 1]$. Based on this score, the confidence classifier $\mathcal{C}(\cdot)$ assigns a discrete label $\gamma(t)$, indicating whether the thought belongs to a low-confidence band (L), a medium-confidence band (M), or a high-confidence band (H). This classification is formally expressed as

$$\gamma(t) = \begin{cases} L, & \text{if } \text{conf}(t) < \alpha, \\ M, & \text{if } \alpha \leq \text{conf}(t) < \beta, \\ H, & \text{if } \text{conf}(t) \geq \beta, \end{cases} \quad \gamma(t) \in \{L, M, H\}, \quad (3.2)$$

where $0 < \alpha < \beta < 1$ are two adjustable thresholds that determine the confidence boundaries. The use of three discrete confidence levels provides a structured way to decide how thoughts should interact during the fusion process.

FoT introduces six types of fusion strategies, determined by the confidence-band assignments of candidate thoughts. These include intra-band strategies (HH, MM, LL) where thoughts of the same confidence level are fused, and cross-band strategies (MH, LH, LM) where thoughts of different confidence levels are combined. Fusion is only triggered when the semantic consistency within a candidate subset is sufficiently high. To regulate this process,

we introduce a round-dependent base threshold τ_t , which increases gradually across reasoning steps, enforcing stricter semantic consistency as reasoning deepens. For each fusion strategy $\phi \in \{HH, MM, LL, MH, LH, LM\}$, a final effective threshold Θ_ϕ is defined as

$$\Theta_\phi = \max(\theta_\phi, \tau_t + \delta_\phi), \quad (3.3)$$

where θ_ϕ is the baseline similarity requirement for strategy ϕ , and δ_ϕ is an additional tightening margin that ensures robustness. This definition guarantees that the actual threshold for fusion is never looser than both the baseline and the dynamically adjusted round-dependent requirement.

Let $\mathcal{T}' \subseteq G_t$ be a subset of candidates selected for possible fusion, with $n = |\mathcal{T}'|$. We define $Z_n = \binom{n}{2}^{-1}$ as a normalization factor. The similarity function $\text{sim}(\cdot, \cdot)$ measures semantic proximity between two thoughts, typically implemented by cosine similarity over embeddings. The average semantic similarity within a candidate subset is given as

$$\text{AvgSim}(\mathcal{T}') = Z_n \sum_{1 \leq i < j \leq n} \text{sim}(t^i, t^j). \quad (3.4)$$

If the condition $\text{AvgSim}(\mathcal{T}') \geq \Theta_\phi$ is satisfied, then the fusion function $F(\cdot)$ merges the subset $\mathcal{T}'$ into a new thought $\tilde{t} = F(\mathcal{T}')$. Let F_t denote the set of all fused thoughts successfully generated at round t . The fused thoughts are then integrated with the original candidates through the update function $U(\cdot)$, yielding the input state for the next reasoning round,

$$S_t = U(G_t, F_t). \quad (3.5)$$

After the iterative reasoning process, the framework must decide which candidate thought is the most reliable. To capture consensus among candidates, we introduce the peer consistency score $\Psi(t)$, which evaluates how semantically aligned a thought t is with the rest of the final thought set S_T . This is defined as

$$\Psi(t) = \frac{1}{|S_T| - 1} \sum_{\substack{u \in S_T \\ u \neq t}} \text{sim}(t, u). \quad (3.6)$$

The peer consistency score encourages the selection of thoughts that not only have high individual confidence but also share semantic agreement with other candidates. Finally, after T rounds of reasoning, the best thought $t^* \in S_T$ is selected according to a joint objective that balances individual confidence and peer support:

$$t^* = \arg \max_{t \in S_T} [\alpha \cdot \text{conf}(t) + \beta \cdot \Psi(t)]. \quad (3.7)$$

The final output of the framework is denoted as $O(S_T)$, where $O(\cdot)$ extracts the task solution from the last-round thought set or directly outputs the response corresponding to t^* .

3.3 Basic idea

The core idea of the Fusion of Thoughts (FoT) framework is to employ a large language model p_θ to iteratively generate intermediate reasoning steps and selectively merge semantically consistent thoughts in order to improve logical coherence and solution quality. The framework operates in a modular fashion, coordinating interactions among prompt construction, semantic parsing, confidence scoring, and fusion control, thereby forming a scalable and flexible reasoning strategy. The reasoning process starts from the task input x . First, the prompt generation module constructs queries suitable for the large language model, and the generator $G(p_\theta, \cdot)$ expands each node in the previous round state S_{t-1} to produce at most k new candidates. The collection of all such candidates constitutes the thought set of round t , denoted as G_t . Each candidate $t \in G_t$ is then evaluated by the scoring module. A normalized confidence score $\text{conf}(t)$ is computed, and through the classifier $\gamma(t)$, each thought is categorized into one of three bands: low (L), medium (M), or high (H). At the same time, the semantic similarity between any two thoughts is measured by the function $\text{sim}(t^i, t^j)$, which provides the basis for determining potential fusions. Under the supervision of the controller,

the framework examines subsets of thoughts $\mathcal{T}' \subseteq G_t$ that correspond to specific confidence configurations, such as purely medium-confidence candidates or cross-band pairs like low-high (LH). For each candidate subset, the average semantic similarity $\text{AvgSim}(\mathcal{T}')$ is calculated. Fusion is triggered only when this value exceeds the strategy-specific threshold Θ_ϕ , where ϕ indicates the chosen fusion strategy. If the condition is satisfied, the fusion function $F(\mathcal{T}')$ produces a new merged thought $\tilde{t} \in F_t$. The newly fused thoughts F_t are then combined with the original candidate set G_t through the update function $U(G_t, F_t)$, forming the state S_t for the next round of reasoning. This iterative process continues until the predefined maximum number of rounds T is reached.

At the final stage, the framework selects the best thought $t^* \in S_T$. The decision is made by balancing the individual confidence score $\text{conf}(t)$ with the peer consistency score $\Psi(t)$, which reflects the semantic agreement of a thought with the rest of the final set. The thought t^* that maximizes this joint objective is then used to derive the final system output, denoted as $O(S_T)$. In summary, the Fusion of Thoughts (FoT) framework provides a principled mechanism for structured reasoning with large language models. By integrating multi-round candidate generation G_t , confidence-based classification $\gamma(t)$, and strategy-specific semantic fusion governed by thresholds Θ_ϕ , FoT achieves a balance between exploratory diversity and semantic consolidation. The incorporation of the peer consistency score $\Psi(t)$ further ensures that the final selection t^* reflects both individual reliability and collective agreement. This combination of components enables FoT to enhance reasoning trajectories in terms of coherence, robustness, and answer quality, offering a generalized methodology that can be flexibly adapted to diverse reasoning tasks and extended to different structural paradigms such as tree and graph-based reasoning.

Chapter 4

The Proposed Fusion of Thought Framework

In this chapter, we present the Fusion of Thoughts (FoT) framework and its two instantiations, ToT^+ and GoT^+ , which extend traditional reasoning architectures with fusion-based control. The flowchart of the proposed framework is shown in figure 4.1.

1. **Phase I: Generic FoT Procedure** — This stage introduces the core operations of FoT: candidate generation with $G(p_\theta, \cdot)$, confidence classification $\gamma(t)$, similarity evaluation $\text{AvgSim}(\cdot)$, fusion under thresholds Θ_ϕ , and state update $U(\cdot)$. These serve as general modules that can be embedded into any reasoning backbone.
2. **Phase II: ToT^+ Construction** — In this stage, the FoT module is integrated into the Tree-of-Thought (ToT) backbone. The result is ToT^+ , which retains the hierarchical expansion and multi-branch search of ToT while embedding FoT’s confidence- and similarity-driven fusion. Fusion is restricted to nodes of the same depth, so the next frontier S_t is updated only from the current layer’s candidates and fused thoughts. This ensures layer-wise structural consistency and controlled exploration.
3. **Phase III: GoT^+ Construction** — In this stage, the FoT module is integrated into the Graph-of-Thought (GoT) framework. The result is GoT^+ , which maintains a global node set V_t and similarity edges E_t , enabling cross-layer fusion inside connected components. By combining GoT’s flexible connectivity with FoT’s fusion criteria, GoT^+

dynamically consolidates information across layers, improving reasoning robustness and efficiency.

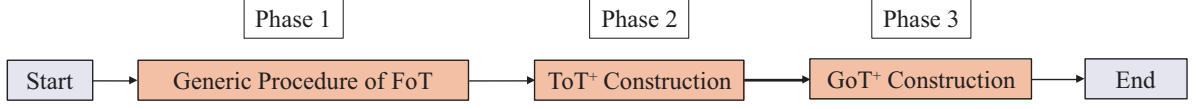


Figure 4.1: The flowchart of the proposed Fusion of Thoughts framework and its ToT⁺ and GoT⁺ instantiations.

Compared with traditional Tree-of-Thought and Graph-of-Thought methods, the FoT-enhanced versions introduce confidence-aware, similarity-driven fusion that filters and merges intermediate thoughts instead of treating them as isolated. This design improves coherence, reduces redundancy, and increases robustness of multi-step reasoning. By explicitly incorporating both semantic similarity and confidence calibration into the reasoning process, FoT provides a principled mechanism for balancing exploration and consolidation. Unlike ToT, which primarily emphasizes breadth through multi-branch expansion, or GoT, which generalizes reasoning into graph structures without fine-grained quality control, FoT continuously evaluates the reliability of candidate thoughts and selectively integrates only those that are semantically aligned and sufficiently confident. This ensures that the reasoning trajectory not only explores diverse paths but also converges toward logically consistent and semantically grounded conclusions. Furthermore, the integration of a fusion mechanism reduces the risk of propagating noisy or contradictory intermediate steps, a common issue in large-scale branching strategies.

4.1 Phase I: Generic Procedure of FoT

The Fusion of Thought (FoT) methodology employs a systematic approach to reasoning, designed to efficiently process and select information. To fully grasp its operation, we can break down its core procedure into distinct, manageable phases. According to Algorithm 1 and as illustrated in figure. 4.2, the general reasoning process of FoT can be roughly divided

into four stages: (1) *reading and initialization*, where the input and context are prepared; (2) *candidate generation and scoring*, where multiple reasoning paths are produced and quantitatively assessed; (3) *subset fusion*, where semantically consistent and confidence-supported thoughts are selectively merged into unified representations; and (4) *update and final selection*, where fused candidates are re-evaluated, iteratively refined, and a final solution is chosen.

This phased procedure highlights the novelty of FoT: it is not merely a static expansion of thoughts but a dynamic pipeline that continually filters, aligns, and consolidates reasoning trajectories. As such, FoT can be viewed as a meta-framework that overlays semantic awareness onto structural prompting, thereby bridging the gap between diversity-driven exploration and coherence-driven consolidation.

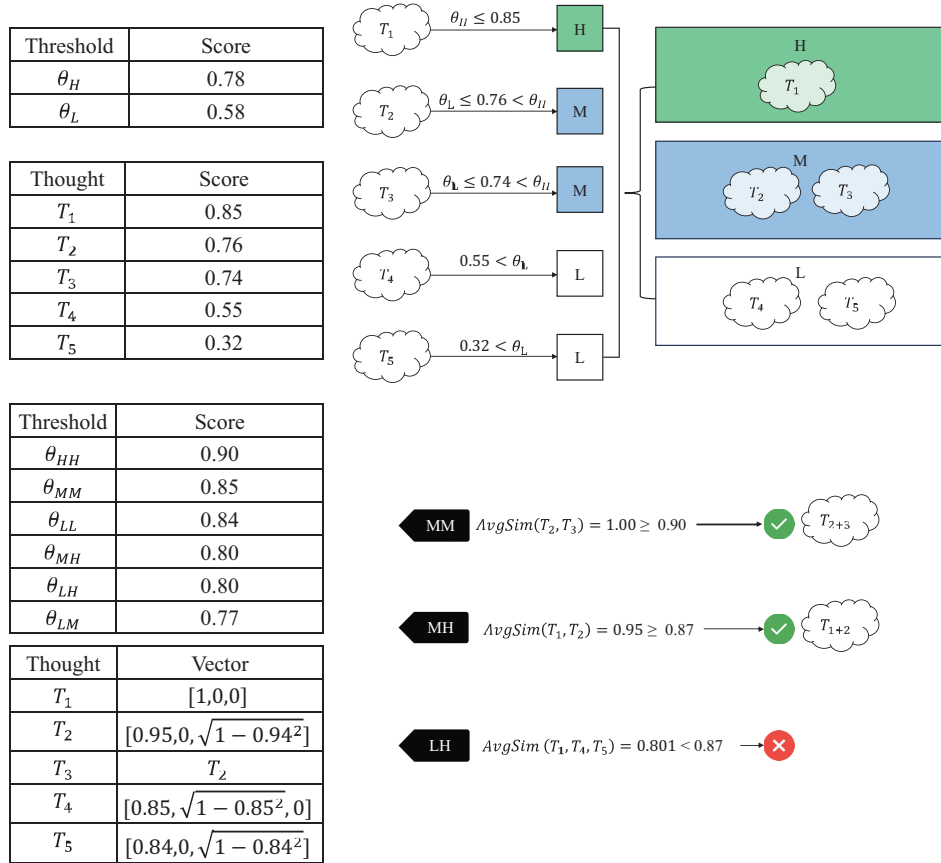


Figure 4.2: Confidence-based fusion strategy, where thoughts in High (H), Medium (M), and Low (L) classes are merged using MM, MH, or LH rules.

- S1.** This block is responsible for handling reading and initialization. First, in lines 1-3, the algorithm reads all necessary parameters, including input node x , prompt p_θ , structure graph G , confidence evaluation function C , similarity function sim , similarity threshold τ , the maximum iteration count T , fusion subset size limit k and target y' , and obtains the pluggable expansion operator Exp and selection operator Sel at the same time. In line 2, it sets the initial thought set $S_0 = \{x\}$, and global record $R \leftarrow S_0$, before entering the main loop from $t = 1$ to T .
- S2.** After entering lines 4-6, the algorithm calls the incoming expansion operator through $G_t = Exp(S_{t-1})$ to generate a batch of new candidate nodes from the previous round of thought set. Then the confidence scores $C(g)$ are calculated for these candidates in line 5 and marked as low (L), medium (M), and high (H) for subsequent selection. Line 6 clears the current round of fusion temporary set F_t and prepares to collect all fusible subsets.
- S3.** Lines 7-10 are the core “subset fusion” stage: the algorithm traverses all subsets $T' \subseteq G_t$ whose size is not less than 2 and corresponds to a specific fusion strategy (e.g., MM, MH, LH). For each subset, the average pairwise similarity is computed as $AvgSim(T') = \frac{2}{n(n-1)} \sum_{i < j} sim(t^i, t^j)$, where $n = |T'|$. If $AvgSim(T') \geq \Theta_\phi$, the fusion function $F(T')$ merges the subset into a single new thought and adds it to F_t .
- S4.** Finally, in lines 11-14, the algorithm calls the selection operator with $S_t = Sel(G_t, F_t)$ to decide which candidates and fusion results to keep in the next round. Then the optimal thought is selected from S_T with $t^* = \arg \max_{u \in S_T} [\alpha C(u) + \beta \Psi(u)]$, and returned as the final output.

Suppose the candidate thoughts are initialized as $T_1 = 0.85$, $T_2 = 0.76$, $T_3 = 0.74$, $T_4 = 0.55$, and $T_5 = 0.32$. Based on the thresholds for individual classification $\theta_L = 0.85$ and $\theta_H = 0.78$, the thoughts are separated into $H = \{T_1\}$, $M = \{T_2, T_3\}$, and $L = \{T_4, T_5\}$. The group fusion thresholds are set as $\Theta_{MM} = 0.90$, $\Theta_{MH} = 0.87$, and $\Theta_{LH} = 0.87$. The

algorithm then evaluates each subset using $AvgSim(T')$. For the medium-confidence group $\{T_2, T_3\}$, $AvgSim = 1.00 \geq \Theta_{MM}$, so they are fused into $\tilde{T}_{2+3}$. For the mixed group $\{T_1, T_2\}$, $AvgSim = 0.95 \geq \Theta_{MH}$, so they are fused into $\tilde{T}_{1+2}$. In contrast, for the low-high group $\{T_1, T_4, T_5\}$, the result is $AvgSim = 0.801 < \Theta_{LH}$, so the fusion is not triggered.

Algorithm 1: FoT Core Procedure

```

1 Initialize parameters:  $x, p_\theta, T, k, \alpha, \beta, \{\tau_t\}_{t=1}^T, \{\theta_\phi\}_\phi, \{\delta_\phi\}_\phi, \text{sim}(\cdot, \cdot)$ .
2 Set  $S_0 = \{x\}$  and  $R \leftarrow S_0$ .
3 for  $t = 1 : T$  do
4   Generate candidates  $G_t = \bigcup_{s \in S_{t-1}} G(p_\theta, s, k)$  (Eq. (3.1)).
5   for each  $g \in G_t$  do
6     Compute  $\text{conf}(g) = V(g)$ ; assign  $\gamma(g) \in \{L, M, H\}$  by Eq. (3.2).
7   end
8    $F_t \leftarrow \emptyset$ .
9   for each strategy  $\phi \in \{HH, MM, LL, MH, LH, LM\}$  do
10     $\Theta_\phi \leftarrow \max(\theta_\phi, \tau_t + \delta_\phi)$  (Eq. (3.3)).
11    for each subset  $\mathcal{T}' \subseteq G_t$  consistent with  $\phi$  do
12      Compute  $AvgSim(\mathcal{T}')$  by Eq. (3.4).
13      if  $AvgSim(\mathcal{T}') \geq \Theta_\phi$  then
14         $F_t \leftarrow F_t \cup \{F(\mathcal{T}')\}$ .
15      end
16    end
17  end
18  Update  $S_t = U(G_t, F_t)$  (Eq. (3.5));  $R \leftarrow R \cup S_t$ .
19 end
20 Compute  $\Psi(u)$  for all  $u \in S_T$  by Eq. (3.6).
21 Select  $t^* = \arg \max_{u \in S_T} [\alpha \cdot \text{conf}(u) + \beta \cdot \Psi(u)]$  (Eq. (3.7)).
22 Output:  $t^*$ .

```

4.2 Phase II: ToT⁺ Construction

In this phase, the Tree-of-Thought with Fusion-of-Thought (ToT⁺) framework is constructed by embedding the FoT mechanism into the traditional ToT structure. The goal is to maintain the strict hierarchical expansion of the tree while selectively merging semantically consistent thoughts at each layer, thereby improving coherence and robustness. The detailed operation of the ToT⁺ architecture is illustrated in Algorithm 2, and an example is shown in figure 4.3.

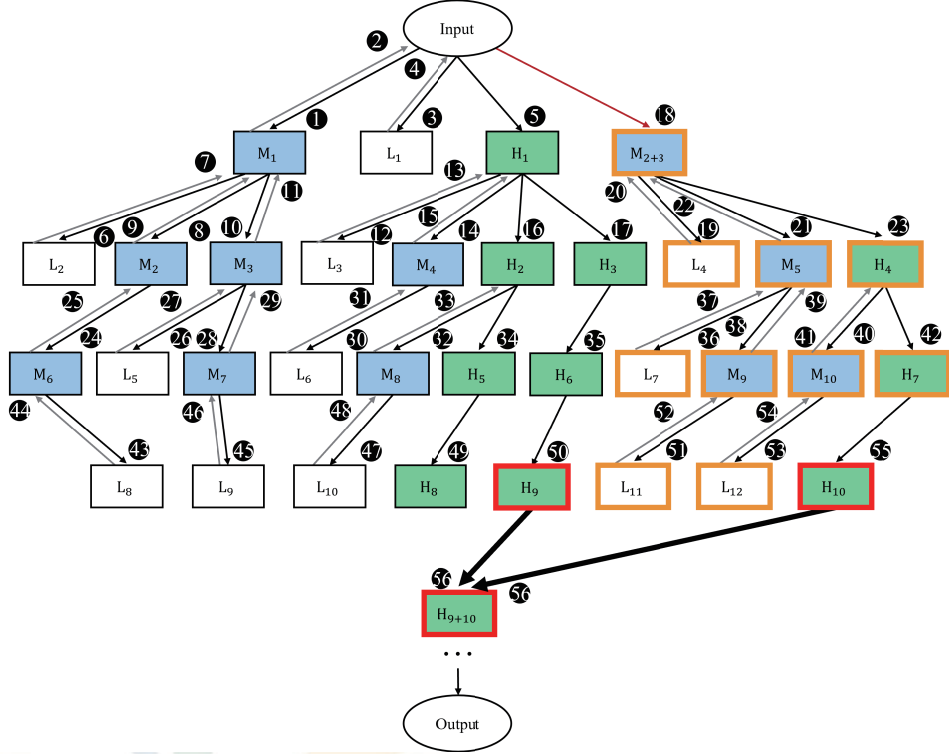


Figure 4.3: Example of FoT reasoning trajectory where multiple candidate thoughts are fused into representative ones to derive the final answer.

- S1.** The algorithm begins at the root node x , where the initial active set is defined as $S_0 = \{x\}$ and the depth of the root is explicitly assigned as $\text{depth}(x) = 0$. At this stage, the global record R is initialized as $R = S_0$, which guarantees that all subsequent layers can be traced back to the initial input. This initialization not only sets the foundation of the tree structure but also establishes the mechanism for recording and accumulating all candidate thoughts across layers. By doing so, the framework ensures that the reasoning process has a well-defined starting point and a global memory for later selection.
- S2.** At each reasoning depth t , the algorithm expands the active set S_{t-1} by generating new candidate thoughts $G_t = \bigcup_{s \in S_{t-1}} G(p_\theta, s, k)$. Each generated candidate $g \in G_t$ is tagged with $\text{depth}(g) = t$, which explicitly binds it to the current layer of the tree. After generation, the confidence score $\text{conf}(g)$ is evaluated using the scoring function $V(\cdot)$, and each candidate is classified into one of three confidence categories $\{L, M, H\}$ according to the thresholds θ_L and θ_H . This explicit labeling is crucial: it provides the structural

information needed for later fusion, and it allows the framework to separate candidates of different quality levels before further processing.

- S3.** Once candidates are generated and classified, the algorithm proceeds to the intra-layer fusion stage. For each fusion strategy $\phi \in \{HH, MM, LL, MH, LH, LM\}$, the effective threshold Θ_ϕ is determined by combining the base threshold θ_ϕ with the adaptive margin from the current round. Subsets $T' \subseteq G_t$ are considered eligible only if all of their members belong to the same depth t and their confidence pattern matches ϕ . For every such subset, the average semantic similarity $AvgSim(T')$ is computed according to Eq. (3.4). If $AvgSim(T') \geq \Theta_\phi$, the subset is regarded as semantically consistent and is fused into a new thought $F(T')$, which is then added to the temporary fusion set F_t . This mechanism ensures that only sufficiently coherent and structurally valid groups of thoughts are merged, thereby improving reasoning quality without introducing noise.
- S4.** After the fusion stage, the system integrates all candidates and fused thoughts by forming the updated set $S_t = U(G_t, F_t)$. This updated set not only contains the original candidates but also the newly fused thoughts that passed the semantic similarity test. The set S_t is then accumulated into the global record $R \leftarrow R \cup S_t$, which stores all reasoning states across layers. The iterative process continues until the maximum tree depth D is reached. At this point, the final decision is made: the optimal thought is selected from the global pool R according to the scoring objective $t^* = \arg \max_{u \in R} [\alpha \cdot C(u) + \beta \cdot \Psi(u)]$. This selection balances both confidence scores and semantic alignment with the task target y' , ensuring that the final output reflects not only locally consistent reasoning but also the best global solution derived from the entire search trajectory.

As illustrated in figure 4.3, the process begins with the root node expanding into the first-layer candidates $\{M_1, L_1, H_1\}$. At depth $t = 2$, nodes $\{L_2, M_2, M_3, L_3, M_4, H_2, H_3\}$ are generated. Within this layer, the medium-confidence pair $\{M_2, M_3\}$ achieves $AvgSim = 1.00 \geq \Theta_{MM}$ and is fused into $\tilde{M}_{2+3}$. At deeper layers, the fused node $\tilde{M}_{2+3}$ expands into

new candidates $\{L_4, M_5, H_4\}$, and at depth $t = 3$, the high-confidence nodes $\{H_9, H_{10}\}$ satisfy $AvgSim \geq \Theta_{HH}$ and are fused into $\tilde{H}_{9+10}$. Together with retained nodes such as H_8 , these candidates form the final pool S_T , from which $\tilde{H}_{9+10}$ is selected as the optimal thought t^* .

Algorithm 2: ToT⁺

```

1 Initialize:  $x, p_\theta, T, k, \alpha, \beta, \{\tau_t\}, \{\theta_\phi\}, \{\delta_\phi\}, \text{sim}$ ; set  $S_0 = \{x\}$ , depth( $x$ ) = 0.
2 for  $t = 1 : T$  do
3    $G_t = \bigcup_{s \in S_{t-1}} G(p_\theta, s, k)$ ; set depth( $g$ ) =  $t$ ,  $\forall g \in G_t$ .
4   for  $g \in G_t$  do
5     compute conf( $g$ ); assign  $\gamma(g)$  by (3.2).
6   end
7    $F_t \leftarrow \emptyset$ .
8   for  $\phi \in \{HH, MM, LL, MH, LH, LM\}$  do
9      $\Theta_\phi \leftarrow \max(\theta_\phi, \tau_t + \delta_\phi)$ .
10    for  $\mathcal{T}' \subseteq G_t$  s.t.  $\forall t^i \in \mathcal{T}' : \text{depth}(t^i) = t$  and labels match  $\phi$  do
11      compute AvgSim( $\mathcal{T}'$ ) by (3.4);
12      if  $AvgSim(\mathcal{T}') \geq \Theta_\phi$  then
13         $F_t \leftarrow F_t \cup \{F(\mathcal{T}')\}$ .
14      end
15    end
16  end
17   $S_t \leftarrow U(G_t, F_t)$  (Eq. (3.5)).
18 end
19 Compute  $\Psi(u)$  for all  $u \in S_T$  by (3.6); select  $t^*$  by (3.7).
20 Output:  $t^*$ .

```

4.3 Phase III: GoT⁺ Construction

In this phase, we introduce the Graph-of-Thought with Fusion-of-Thought (GoT⁺) architecture, which extends the FoT framework into a flexible graph-based reasoning process. Unlike ToT⁺, which enforces a strict tree structure with depth-based expansion, GoT⁺ leverages graph topology to enable iterative expansion, cross-layer connections, and multi-round fusion. This flexibility allows GoT⁺ to revisit and refine promising reasoning paths, achieving broader search coverage and repeated optimization. The detailed operation of GoT⁺ is illustrated in Algorithm 3.

S1. The algorithm begins by setting the root input x as the initial active set $S_0 = \{x\}$ with $V_0 \leftarrow S_0$ and $E_0 = \emptyset$, which initializes both the vertex and edge sets of the reasoning

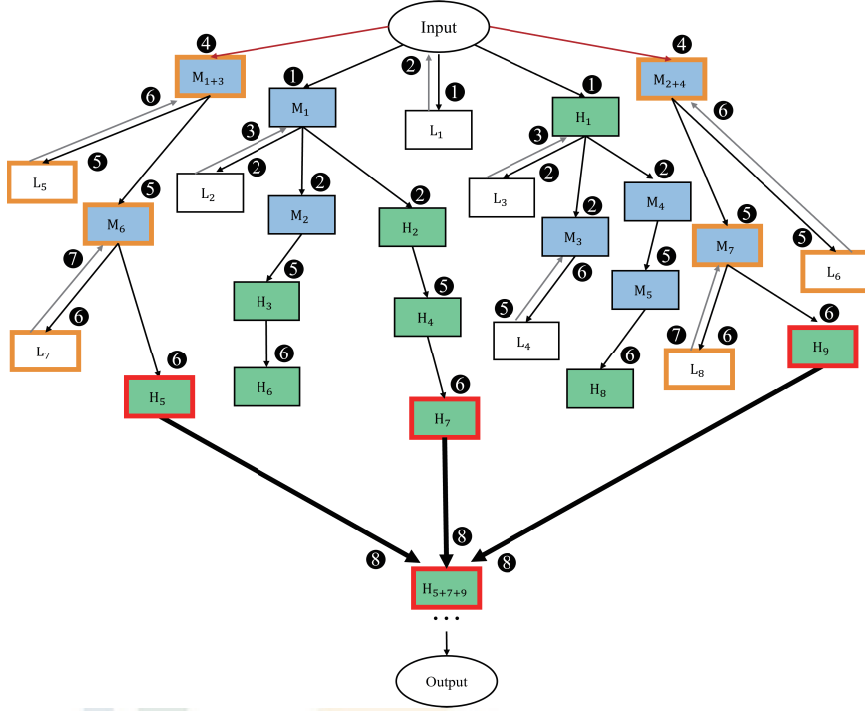


Figure 4.4: GoT⁺ Initial candidate thoughts are generated from the input, then integrated with GoT candidate thoughts through the 4.1 FoT architecture to form new thoughts.

graph. This guarantees that the graph starts with a single input node but is ready to accumulate connections as expansion proceeds.

- S2.** At each iteration t , new candidates $G_t = \bigcup_{s \in S_{t-1}} G(p_\theta, s, k)$ are generated and integrated into the global vertex set $V_t \leftarrow V_{t-1} \cup G_t$. Each $g \in G_t$ is scored with the confidence function $\text{conf}(g)$ and classified into $\{L, M, H\}$ categories for subsequent fusion. Semantic similarity between all pairs (i, j) of nodes in V_t is computed, and pairs exceeding the similarity threshold τ_t are stored as edges E_t .
- S3.** Candidate communities C_t are extracted from (V_t, E_t) using connected components. For each fusion strategy $\phi \in \{HH, MM, LL, MH, LH, LM\}$, the effective threshold Θ_ϕ is set, and subsets $T' \subseteq C$ consistent with the strategy are considered. If the average similarity $\text{AvgSim}(T') \geq \Theta_\phi$, then T' is fused into a new thought $F(T')$ and added to the fusion set F_t . This stage enables GoT⁺ to merge across broader structural contexts rather than restricting fusion within a single depth.

- S4.** The merged set $S_t = U(G_t, F_t)$ is added back into the vertex set $V_t \leftarrow V_t \cup F_t$, ensuring that fused nodes can participate in subsequent rounds of expansion. This mechanism allows GoT⁺ to re-activate and refine previously generated reasoning fragments, thereby improving depth and consistency across multiple iterations.
- S5.** After T rounds, the accumulated candidate pool S_T is evaluated globally. The optimal thought is selected as $t^* = \arg \max_{u \in S_T} [\alpha \cdot C(u) + \beta \cdot \Psi(u)]$, balancing confidence scores and semantic similarity with respect to the target.

As illustrated in figure 4.4, the process begins with the input node expanding into the first-layer candidates $\{L_1, H_1, L_2\}$. In subsequent rounds, additional nodes are generated, such as $M_1 \rightarrow L_2$, $L_3 \rightarrow H_1$, and M_4 branching from another path. At depth $t = 2$, semantic similarity is computed among all second-layer candidates. In the *MM* strategy, nodes $\{M_2, M_4\}$ satisfy $\text{AvgSim} \geq \Theta_{MM}$ and are fused into $\tilde{M}_{2+4}$ (orange box). Similarly, $\{M_3, M_5\}$ are fused under the same condition. In the *LH* strategy, pairs $\{L_2, H_3\}$ and $\{L_4, H_2\}$ exceed Θ_{LH} and are merged. These fused nodes, such as $\tilde{M}_{2+4}$, continue to expand into deeper layers, producing nodes like $\{M_6, L_5, H_4\}$ at step 8. In the third layer, high-confidence nodes $\{H_5, H_7, H_9\}$ (red boxes) exhibit strong similarity and are fused into $\tilde{H}_{5+7+9}$. Along with retained nodes such as H_6 and H_{10} , these candidates form the final pool S_T . Ultimately, FoT’s selection criterion identifies $\tilde{H}_{5+7+9}$ as the optimal thought t^* , which is used to generate the final output.

Algorithm 3: GoT⁺

```
1 Initialize:  $x, p_\theta, T, k, \alpha, \beta, \{\tau_t\}, \{\theta_\phi\}, \{\delta_\phi\}, \text{sim}$ ; set  $S_0 = \{x\}$ ,  $V_0 \leftarrow S_0$ ,  $E_0 = \emptyset$ .
2 for  $t = 1 : T$  do
3    $G_t = \bigcup_{s \in S_{t-1}} G(p_\theta, s, k)$ ;  $V_t \leftarrow V_{t-1} \cup G_t$ .
4   for  $g \in G_t$  do
5     | compute  $\text{conf}(g)$ ; assign  $\gamma(g)$  by (3.2).
6   end
7    $E_t \leftarrow \{(i, j) \mid i \neq j, i, j \in V_t, \text{sim}(i, j) \geq \tau_t\}$ .
8    $F_t \leftarrow \emptyset$ ; let  $\mathcal{C}_t = \mathbf{CC}(V_t, E_t)$ .
9   for  $\phi \in \{HH, MM, LL, MH, LH, LM\}$  do
10     $\Theta_\phi \leftarrow \max(\theta_\phi, \tau_t + \delta_\phi)$ .
11    for  $C \in \mathcal{C}_t$  do
12      | for  $\mathcal{T}' \subseteq C$  s.t. labels match  $\phi$  do
13        | compute  $\text{AvgSim}(\mathcal{T}')$ ; if  $\text{AvgSim}(\mathcal{T}') \geq \Theta_\phi$  then
14          |  $F_t \leftarrow F_t \cup \{F(\mathcal{T}')\}$ .
15        | end
16      | end
17    end
18  end
19   $S_t \leftarrow U(G_t, F_t)$ ;  $V_t \leftarrow V_t \cup F_t$ .
20 end
21 Compute  $\Psi(u)$  for all  $u \in S_T$  by (3.6); select  $t^*$  by (3.7).
22 Output:  $t^*$ .
```

Chapter 5

Experimental Result

In this chapter, the experimental setup is explained in section 5.1, the performance evaluation is discussed in section 5.2.

5.1 Experimental setup

In the thesis, we aim to conduct an in-depth investigation and comparison of large language models' (LLMs) performance on question answering tasks under various reasoning paradigms. Grounded in the Fusion of Thoughts (FoT) framework, we have implemented a spectrum of reasoning strategies ranging from basic to advanced. These include the most direct Direct I/O (IO), the step-by-step Chain of Thought (CoT), the classical Tree-of-Thoughts (ToT), and the more flexible Graph-of-Thoughts (GoT). To further bolster reasoning capabilities, we introduce two enhanced variants: thought fusion-based Tree-of-Thoughts (ToT⁺) and thought fusion-based Graph of Thoughts (GoT⁺). In this section, we elaborate on the experimental design, core components, and parameter configurations for each of these methods.

5.1.1 Dataset

In this thesis, we selected TriviaQA as our primary dataset. TriviaQA is a high-quality English question answering corpus renowned for its broad coverage of knowledge domains and diverse inference difficulties. Its questions are directly sourced from fourteen real-world Q&A and quiz websites; These items, typically authored by trivia enthusiasts, exhibit significant

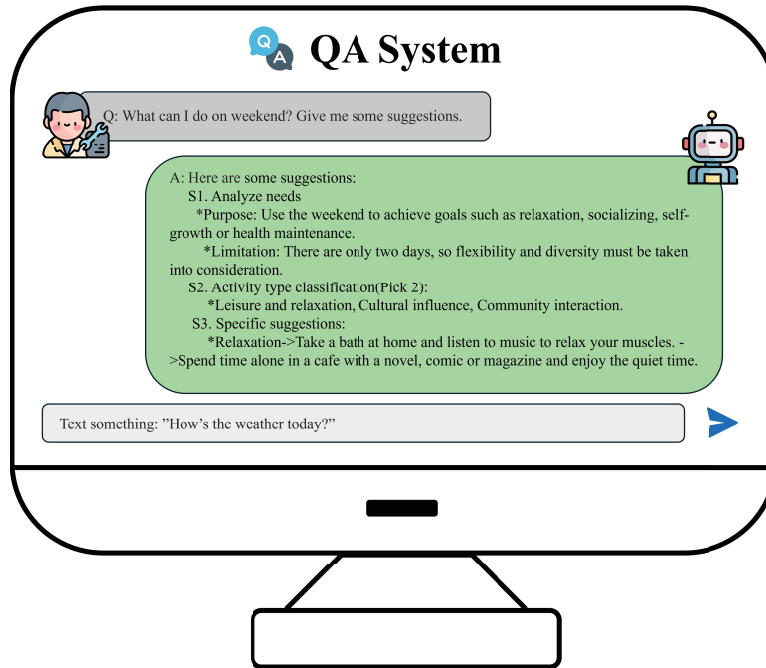


Figure 5.1: The QA system’s query interface shows that, when a user submits a question, the system employs a multi step reasoning process to iteratively generate and present the response that best addresses the query.

real-world complexity and variety. They span history, science, culture, and the arts and often require multi-step reasoning to answer. The corresponding evidence documents are collected independently of the question writing process, drawing on Wikipedia and a wide range of web resources. This distantly supervised construction ensures a challenging relationship between questions and evidence, since answers may not appear in a single sentence or paragraph but must instead be integrated and inferred from multiple sources.

To comprehensively assess the model’s diverse capabilities, we applied two specific pre-processing treatments to the TriviaQA dataset and produced the corresponding data files: Open-Domain QA evaluates the model’s ability to interpret and leverage external supporting information. In addition to the basic question and its human-authored answer, we provide an aggregated context in the contexts field. These contexts are obtained by retrieving the top five documents for each TriviaQA query and concatenating their contents. By pre-integrating this background information, the model can directly draw on rich supporting evidence during few-shot learning, allowing us to assess its reasoning performance when given explicit external

knowledge. Open-Ended QA isolates the model’s intrinsic reasoning capabilities by removing any external context. Each instance contains only the question and the single answer ; the contexts field is omitted entirely. This setup ensures that the model cannot rely on retrieved evidence and must depend solely on its internal knowledge and logical inference to produce the most plausible answer. It particularly challenges the model’s multi-step logical composition and semantic alignment, since answers must be derived “from scratch” rather than extracted from provided texts.

5.1.2 FoT QA System Architecture

We built an open-domain question answering system atop the FoT framework, in which multiple modules collaborate to realize a complex reasoning pipeline. First, the prompt generator dynamically constructs an LLM prompt that includes the question, relevant context, and a strategy template in accordance with the requirements of the current FoT operation. It then augments the prompt with requisite few-shot examples to enhance generation quality. Next, the large language model (LLM; OpenAI GPT-3.5 Turbo) concurrently produces several preliminary reasoning threads upon receiving this prompt, with its parameters (temperature = 1.0, max_tokens = 1536) ensuring both diversity and brevity. The output parser then cleans and extracts the raw textual outputs, converting them into structured state objects for uniform downstream processing.

The controller serves as the pipeline’s central orchestrator and drives each node’s execution according to the predefined Operation Graph, which specifies the sequencing and parallelism of Generate, Score, FuseThoughts, KeepBestN, and other operations. It simultaneously monitors API-call costs, logs activity, and manages the inheritance and fusion of multi-layer reasoning threads. During the FuseThoughts phase, the system computes semantic similarities between threads and merges those exceeding a threshold so as to preserve diversity while eliminating redundancy. Finally, after retaining the optimal solution, the controller invokes the GroundTruth validation or scoring module to produce the final answer. Their interaction flow ensures efficiency, controllability, and consistency across all stages, from question retrieval and

prompt construction, through reasoning expansion and fusion, to answer parsing and output. The detailed structure of these Operation Graphs is presented in Table 5.1.

Table 5.1: Important parameters required for experiment execution.

Main	Parameter
dataset	fewdoc, open
data_ids	0–29
methods	"io", "cot", "tot", "got", "tot_plus", "got_plus"
budget_per_method	50
lm_name	"chatgpt"
index_path	None

As the baseline, the method follows the most direct reasoning path: a single generation step produces one final answer, with no intermediate thought generation or complex multi-step reasoning, and the output is then compared against the ground truth. Chain-of-Thought (CoT) extends the baseline by prompting step-by-step reasoning; although it still performs only one generation step, the prompt explicitly instructs the model to “think step by step” before giving the final answer, with the goal of improving accuracy. Tree-of-Thoughts (ToT) simulates a tree-like search. At the first layer the model generates ten preliminary candidate thoughts, scores them independently, and retains the highest-scoring one. In each of the next two layers, this surviving candidate is expanded into four new candidates, which are again scored independently and pruned to a single best option, yielding a progressively convergent trajectory. Graph-of-Thoughts (GoT) employs a more complex graph-based process. It begins with a planning phase that produces five alternative problem-solving strategies or answer outlines; in the first stage each plan is expanded into four candidates, the candidates are scored independently, and the two highest-scoring candidates are retained overall. In the second stage, these two survivors are each expanded into four candidates; after another round of independent scoring, a final selection step retains a single optimal answer, combining breadth with depth.

ToT⁺ enhances ToT by introducing a FuseThoughts (FoT) module (figure 5.2). For a controlled comparison, its generation and selection budgets are kept identical to those of ToT.

Tree-of-Thought⁺(ToT⁺)

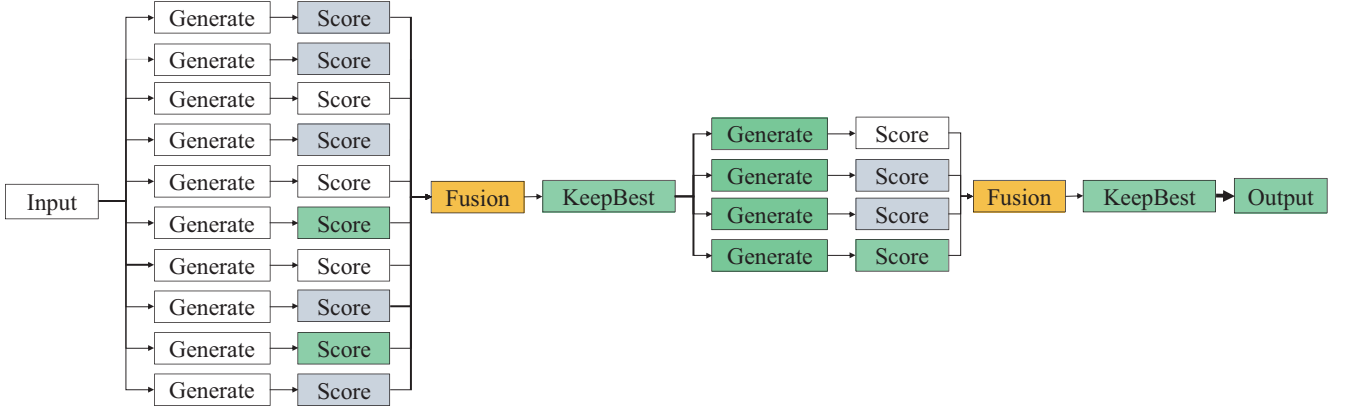


Figure 5.2: This figure illustrates the ToT⁺ inference method’s workflow for QA tasks. Compared to ToT, ToT introduces fusion as a crucial step for thought integration.

At the first layer, the system generates ten candidate thoughts, and the subsequent scoring step evaluates the entire set jointly (combined scoring). FuseThoughts then groups semantically similar candidates using a predefined similarity threshold and a diversity parameter (see Table 1) and merges each group by retaining the longest non-empty candidate as the representative to maximize information retention. After fusion, the procedure retains the single best candidate. The second layer repeats this process with four newly generated thoughts, followed by joint scoring, fusion, and selection of the best candidate.

GoT⁺ deeply integrates thought fusion into GoT’s multi-stage, parallel reasoning process (Figure 5.3). To preserve experimental control, it retains the same settings for Generate and KeepBestN as in GoT. During the planning phase (Phase 0), the system produces five candidate plans. In Phase 1, each plan is expanded into four candidate thoughts, the candidates are batch-scored, and the first FuseThoughts step is applied with the parameterization summarized in Table 1, using the merge-by-longest rule. After fusion, the top two thoughts are retained. In Phase 2, these two thoughts are each expanded into four new candidates, followed by batch scoring and a second FuseThoughts step. Finally, KeepBestN selects a single answer. This staged design enables GoT⁺ to manage parallel thought streams more effectively, reduce redundancy, and extract higher-quality information, thereby improving the efficiency

Graph-of-Thought⁺ (GoT⁺)

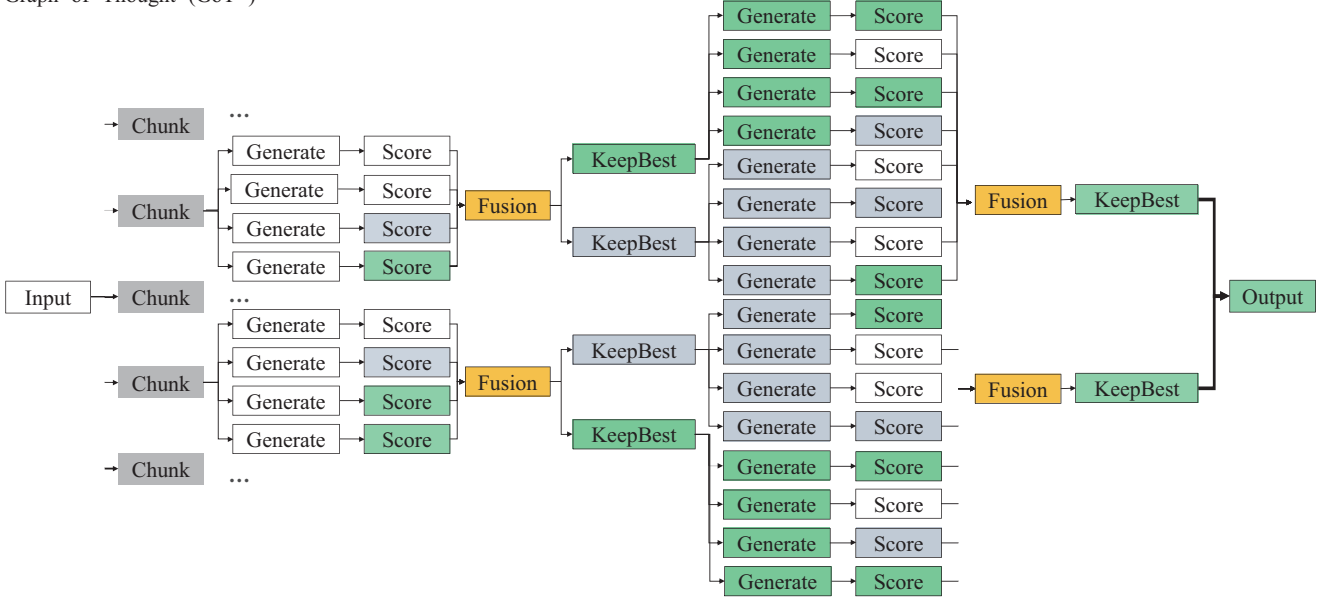


Figure 5.3: GoT⁺ inference method’s workflow for QA tasks. Compared to GoT, GoT⁺ introduces Fusion as a crucial step for thought integration. Additionally, when compared to ToT⁺, its architecture becomes relatively more complex due to the initial embedding of inputs.

of complex reasoning tasks.

5.1.3 Illustrative Example

As illustrated in figure 5.4, the ToT⁺ method processes the question “Who was the man behind The Chipmunks?” through a structured reasoning pipeline rather than answering it directly. In the first phase, multiple candidate thoughts are generated, such as partial clues (e.g., “a songwriter named Dave”) or more complete statements (e.g., “Dave Seville was the man behind the Chipmunks”), which are then classified into confidence bands $\{L, M, H\}$. During the fusion stage, high-confidence candidates referring to “Dave Seville” are found to exceed the similarity threshold Θ_{HH} and are merged into a consolidated thought $\tilde{H}_f$, thereby reducing redundancy while preserving semantic consistency. Finally, the controller applies KeepBestN to select the most representative thought from the fused pool, yielding the final prediction “David Seville”. This example highlights how ToT⁺ decomposes a complex query into intermediate reasoning steps, applies confidence-based grouping and fusion to refine the

candidate set, and ultimately converges on a concise and accurate answer.

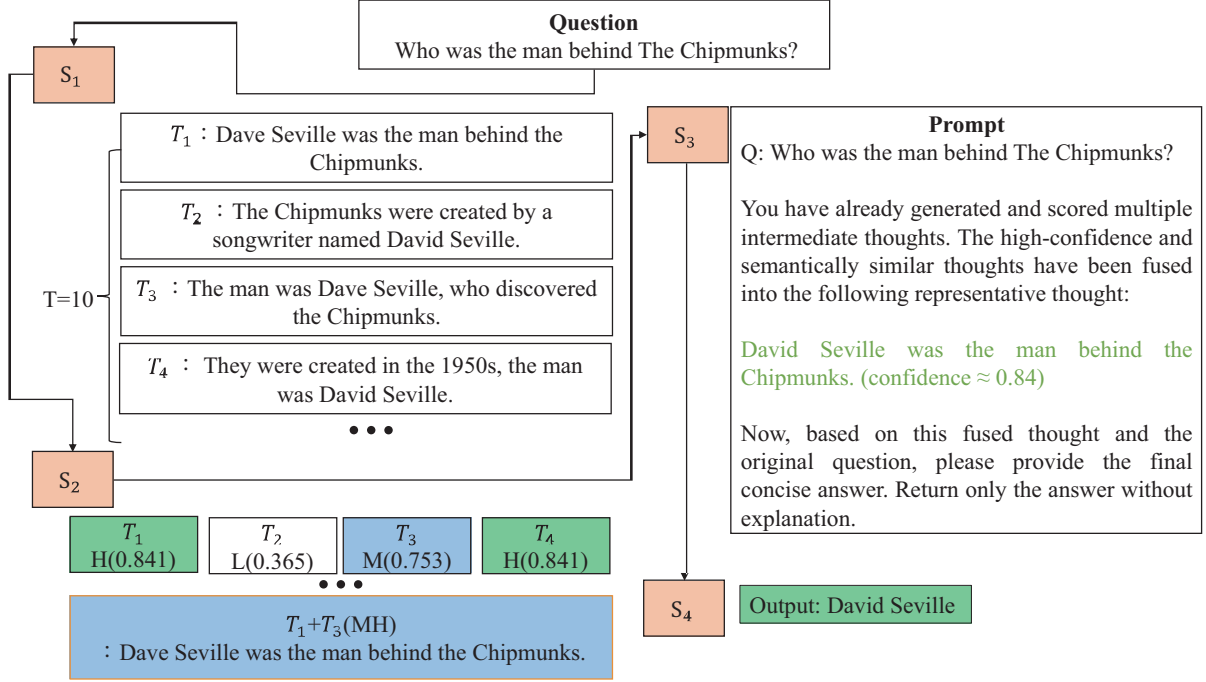


Figure 5.4: Illustration of ToT⁺ reasoning in QA, where semantically similar branches are fused into coherent representative thoughts.

5.2 Performance Analysis

This chapter presents a performance analysis of different reasoning methods based on the numerical results of our experiments. The primary objective of this thesis is to evaluate the reasoning capabilities of Large Language Models (LLMs) when performing Open-Domain Question Answering (Open-Domain QA) and Open-Ended Question Answering (Open-Ended QA) tasks. By comparing various reasoning methods, including baseline methods IO and CoT, Tree-of-Thought (ToT) and its fusion-of-thought (FoT) variant ToT⁺, and Graph-of-Thought (GoT) and its FoT variant GoT⁺, we aim to verify whether the Fusion-of-Thought (FoT) mechanism can effectively enhance the similarity, precision, completeness, semantic quality, as well as the breadth and depth of generated answers. We anticipate the final comprehensive performance ranking to be: GoT⁺, ToT⁺, GoT, ToT, CoT, IO.

5.2.1 Evaluation Methods and Metrics

To comprehensively assess the quality of the answers generated by different reasoning methods, we employed a set of well-established objective metrics and one subjective evaluation metric. Prior to evaluation, all answers underwent preprocessing, including the removal of HTML/XML tags and introductory phrases (e.g., “The answer is:”), extraction of the first sentence as the main answer, and conversion to Title Case for consistency. The objective metrics include F1-score, Exact Match (EM), BLEU, ROUGE, METEOR, and BERTScore. These metrics respectively capture token-level accuracy and completeness, exact lexical match, n -gram fluency, recall-oriented overlap, synonym- and stem-aware matching, and semantic similarity based on contextual embeddings. Since their computational definitions are well established in the literature, we summarize their key properties and corresponding formulas in table 5.2. In addition to the above automatic metrics, we introduced a subjective evaluation metric termed openScore. This metric is designed to approximate human judgment using the LLM itself as an evaluator.

Specifically, each generated answer was re-scored by the ChatGPT model under a carefully constructed prompt that instructed the model to act as a critical reviewer and assign scores across five dimensions: Relevance (R), Completeness (C), Coherence (Co), Depth (D), and Overall Quality (Q). Each dimension was rated on a 1–5 scale (1 = very poor, 5 = excellent), and short textual justifications were also required. The following is an example of the evaluation prompt: ”You are an expert reviewer. Please evaluate the following answer to a question along five dimensions: (1) Relevance, (2) Completeness, (3) Coherence, (4) Depth, and (5) Overall Quality. Each score should be an integer from 1 (very poor) to 5 (excellent). Provide the scores in strict JSON format with keys R, C, Co, D, Q, and also provide a one-sentence justification.” The returned scores were parsed, validated, and then normalized to the [0,1] interval, where a raw score $s_i \in [1, 5]$ was transformed as:

$$n_i = \frac{s_i - 1}{4}. \quad (5.5)$$

The final OpenScore was obtained as the mean of the normalized dimensions:

$$OpenScore = \frac{n_R + n_C + n_{Co} + n_D + n_Q}{5}. \quad (5.6)$$

For example, if a response received ratings $R = 4$, $C = 3$, $Co = 5$, $D = 4$, and $Q = 3$, the normalized values would be 0.75, 0.50, 1.00, 0.75, and 0.50, respectively, yielding a final OpenScore of 0.70. This design allows OpenScore to capture nuanced aspects of answer quality that purely lexical metrics cannot evaluate, especially for open-ended and semantically diverse answers.

Table 5.2: Summary of objective evaluation metrics used in this thesis, with definitions and formulas.

Metric	Definition	Formula
F1-score	Token-level overlap combining precision (P) and recall (R).	Eq. (5.1)
Exact Match (EM)	Binary judgment of whether the predicted answer exactly matches the gold answer.	–
BLEU	n -gram precision with brevity penalty, measuring fluency and word order similarity.	Eq. (5.2)
ROUGE-1/2/L	Recall-oriented overlap metrics: unigram, bigram, and longest common subsequence.	–
METEOR	Stemming/synonym-aware alignment with fragmentation penalty.	Eq. (5.3)–(5.4)
BERTScore	Embedding-based metric computing contextual semantic similarity.	–

5.2.2 Evaluation Results Analysis

This thesis employs various evaluation metrics to comprehensively analyze the performance of different reasoning methods in question-answering tasks. The results are illustrated in figure 5.5, which presents nine evaluation metrics across different sample sizes of randomly selected questions. In this setting, the x-axis represents the number of randomly sampled questions, while the y-axis reports the average value of the respective evaluation metric over those questions.

Compared to the previous cumulative averaging approach, this presentation directly reflects the mean performance at each sampling size, providing a clearer view of stability and scalability as the sample size increases. GoT⁺ (red dashed circles) and ToT⁺ (blue dashed triangles) consistently lead across almost all metrics, significantly surpassing their non-augmented counterparts (GoT and ToT) as well as baseline methods (CoT and IO). As the number of sampled questions increases, the performance curves of GoT⁺ and ToT⁺ not only achieve higher peak values but also exhibit smoother and more stable trajectories, highlighting their advantages in robustness and generalizability. F1-score reflects token-level accuracy and completeness. GoT⁺ and ToT⁺ maintained the highest average F1 values (0.4–0.55 range), clearly outperforming ToT and GoT. CoT remained consistently lower, while IO lagged significantly behind. Exact Match (EM) captures verbatim precision. Again, GoT⁺ and ToT⁺ achieved superior EM rates (around 0.3–0.4), while ToT and GoT performed moderately. CoT and IO consistently stayed below 0.15, indicating their difficulty in producing exact matches. BLEU values show that GoT⁺ and ToT⁺ consistently achieved the best fluency and word-order alignment, stabilizing around 0.08–0.12. By contrast, CoT and IO barely reached 0.05. ROUGE-1, ROUGE-2, and ROUGE-L reveal similar trends: GoT⁺ and ToT⁺ exhibit strong unigram, bigram, and subsequence overlaps (ROUGE-1 around 0.45–0.55, ROUGE-2 around 0.25–0.35, ROUGE-L around 0.6–0.7). Traditional ToT and GoT performed decently but fell below their enhanced versions, while CoT and IO again scored significantly lower. METEOR emphasizes flexible lexical alignment and synonym handling. GoT⁺ and ToT⁺ achieved the highest averages (0.4–0.55), with GoT⁺ occasionally outperforming ToT⁺. CoT and IO remained at 0.2–0.3. BERTScore, which measures semantic similarity, produced higher absolute values across all methods. GoT⁺ and ToT⁺ led slightly (0.8–0.85 range), followed closely by GoT and ToT. CoT and IO, although lower (~0.65–0.75), still achieved moderately high values due to BERTScore’s robustness to paraphrasing. OpenScore, the LLM-based subjective evaluation, demonstrated consistent preference for GoT⁺ and ToT⁺, which both stabilized above 0.85 as the number of sampled questions increased. GoT and ToT followed, while CoT hovered be-

tween 0.65 and 0.7. IO exhibited the lowest performance, around 0.6, with visible instability in smaller sample sizes. This reinforces the conclusion that FoT-enhanced methods generate answers judged by LLMs to be more relevant, complete, coherent, and of higher overall quality.

In summary, across all nine evaluation metrics, GoT⁺ and ToT⁺ consistently outperformed their original versions and baseline methods, validating the effectiveness of the Fusion-of-Thought mechanism in improving both objective accuracy and subjective quality of answers.

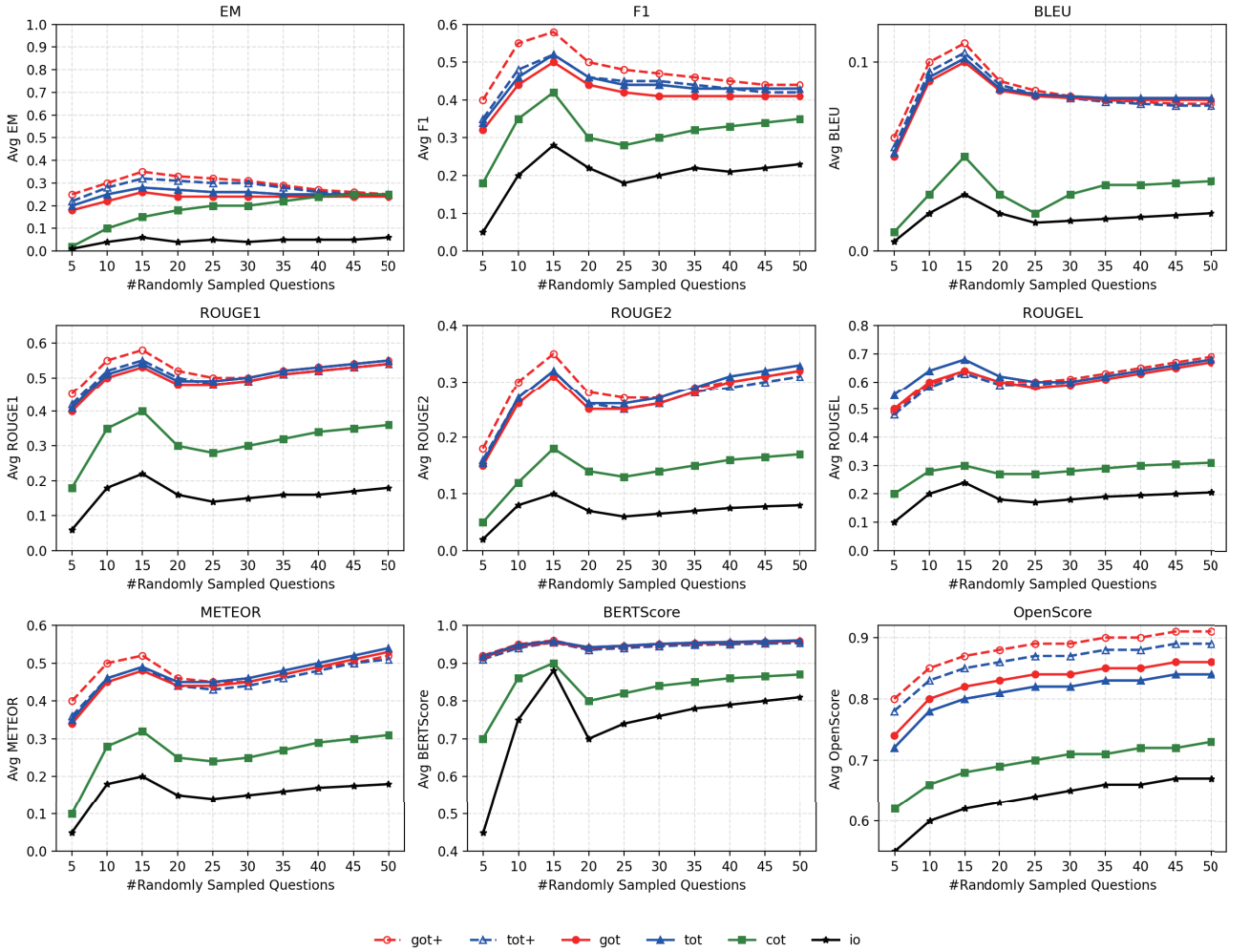


Figure 5.5: 9 evaluation metrics. GoT⁺ (red hollow circles) and ToT⁺ (blue hollow circles) consistently lead, with their cumulative average values significantly surpassing traditional ToT, GoT, CoT, and IO methods. As the sample size increases, the Plus series not only achieves higher peak values but also exhibits smoother and more stable curves, demonstrating their significant advantages in accuracy, diversity, and consistency.

5.2.3 Cost and Token Analysis

Beyond accuracy-oriented metrics, it is equally important to analyze the computational cost and token usage of different reasoning methods, as illustrated in figure 5.6. This analysis provides insights into the efficiency trade-offs associated with adopting more advanced reasoning strategies. The figure reports four complementary perspectives: (a) average cost per question, (b) total cost per run, (c) average prompt tokens per question, and (d) average completion tokens per question.

The top-left panel of figure 5.6 shows that the average cost per question remains almost constant across different sample sizes for all methods. Among them, GoT⁺ is the most expensive (approximately 0.005 per query), followed by GoT (~ 0.004), ToT⁺ (~ 0.0035), and ToT (~ 0.003). CoT and IO, in contrast, are significantly cheaper, at around 0.002 and 0.001, respectively. These results highlight the marginal cost overhead incurred by tree- and graph-based reasoning with semantic fusion. The top-right panel indicates that the total cost per run decreases as the number of sampled questions increases. This is because the overall budget is distributed over more queries, resulting in a lower cost per run at higher sample sizes. Nevertheless, GoT⁺ and GoT consistently consume the highest total cost, while CoT and IO remain the most economical. Bottom-left panel reveals substantial differences in input length across methods. GoT⁺ requires the most extensive prompts, exceeding 3000 tokens per query, while ToT⁺ averages close to 2000 tokens. GoT and ToT consume fewer prompt tokens (around 1700 and 1500, respectively). In comparison, CoT and IO remain lightweight, with input lengths generally below 500 tokens. These results confirm that graph- and tree-based prompting schemes incur significant structural overhead. The bottom-right panel demonstrates a similar pattern in output length. GoT⁺ generates the longest responses, averaging more than 600 tokens per question, followed by ToT⁺ and GoT (200-400 tokens). ToT produces moderately sized completions, while CoT and IO consistently yield concise outputs (typically under 100 tokens).

Overall, these results highlight a clear trade-off between performance and efficiency. While

FoT-enhanced methods (GoT⁺ and ToT⁺) achieve superior accuracy and semantic quality, they do so at the expense of higher computational cost and token consumption. Conversely, lightweight baselines such as CoT and IO are highly cost-efficient but significantly underperform in terms of accuracy and robustness. This observation underscores the importance of balancing reasoning quality with computational efficiency when selecting or designing reasoning frameworks.

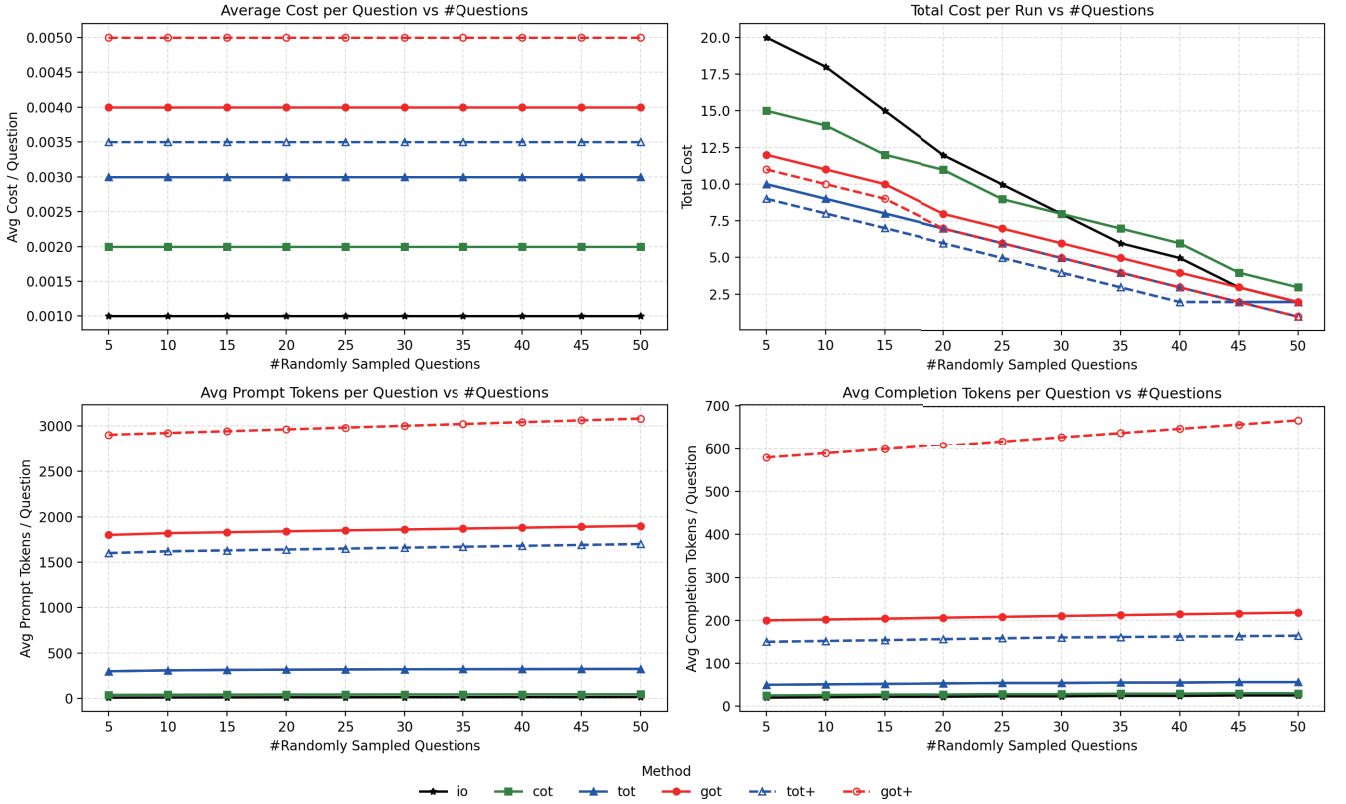


Figure 5.6: The graphs show that GoT⁺ and ToT⁺ have significantly higher average costs and token consumption per question than other methods. However, the total cost for a complete run decreases as the number of questions increases, suggesting a potential for efficiency benefits with larger sample sizes.

5.3 System Implementation

To evaluate the practical applicability of the FoT framework, we designed and implemented an interactive web-based QA system interface (as shown in figure 5.7). In this in-

interface, users can input natural language questions through the web front-end and select any reasoning method to execute the task. The system then automatically launches the reasoning process and displays, on the right panel, the sorted intermediate thoughts, fusion records, and final answers. The interface further supports node visualization, fusion path highlighting, and real-time comparison across methods. It clearly distinguishes between intermediate thoughts and fused thoughts at each stage, thereby helping users intuitively understand the reasoning structure and the conditions for semantic integration. This system serves not only as an experimental platform but also as a prototype reference for future human-computer interaction applications. By visualizing the reasoning trajectory, users are able to inspect the logical structure and semantic evolution of model outputs, which enhances both trust and interpretability of the model's decision process. Moreover, the plug-and-play modular design, supporting multiple reasoning strategies, provides strong extensibility, making it suitable for future integration with knowledge retrieval, context enhancement, and other auxiliary modules. This paves the way for applying reasoning technologies in scenarios such as open-domain QA, educational support, and decision assistance.

Fusion-of-Thought 問答系統

使用 ToT+ 或 GoT+ 進行問答，並依據字數簡易排序。

請輸入問題

想放鬆

選擇推理方法

GoT+

Clear Submit

思路輸出 (依分數排序)

[1] 建議可以找一個安靜舒適的地方，深呼吸幾次，伸展一下身體，聽一些輕柔的音樂或冥想幾分鐘，放鬆身心。也可以進行一些輕鬆的運動或按摩放鬆疲憊的肌肉。另外，和親友聊聊天、閱讀一本好書，或者喝一杯香氣撲鼻的茶也是很好的放鬆方式。無論是什麼方式，重要的是給自己一些時間放鬆和康復。

[2] 可以試著做一些深呼吸運動，或是去散步、聽音樂、看書等平靜放鬆的活動，也可以考慮尋求心理輔導或按摩等專業幫助。重要的是給自己一些時間去照顧自己，讓身心得到釋放和放鬆。

[3] 你可以嘗試做一些舒緩壓力的活動，比如散步、聽音樂、看書、冥想、做瑜伽或者泡一杯熱茶，找到適合自己的放鬆方式，讓自己擺脫壓力和煩惱，放鬆身心。

FUSED Thoughts:

[FUSED 1] 建議可以找一個安靜舒適的地方，深呼吸幾次，伸展一下身體，聽一些輕柔的音樂或冥想幾分鐘，放鬆身心。也可以進行一些輕鬆的運動或按摩放鬆疲憊的肌肉。另外，和親友聊聊天、閱讀一本好書，或者喝一杯香氣撲鼻的茶也是很好的放鬆方式。無論是什麼方式，重要的是給自己一些時間放鬆和康復；可以試著做一些深呼吸運動，或是去散步、聽音樂、看書等平靜放鬆的活動，也可以考慮尋求心理輔導或按摩等專業幫助。重要的是給自己一些時間去照顧自己，讓身心得到釋放和放鬆。

[FUSED 2] 你可以嘗試做一些舒緩壓力的活動，比如散步、聽音樂、看書、冥想、做瑜伽或者泡一杯熱茶，找到適合自己的放鬆方式，讓自己擺脫壓力和煩惱，放鬆身心。

Flag

Figure 5.7: QA system UI.

Chapter 6

Conclusion

In the thesis, we proposed the Fusion-of-Thought (FoT) framework as a general-purpose augmentation mechanism for enhancing the structured reasoning capabilities of Large Language Models (LLMs). By embedding semantic fusion and confidence-based control into existing architectures such as Tree-of-Thought (ToT) and Graph-of-Thought (GoT), we derived the enhanced variants ToT⁺ and GoT⁺. Through extensive experiments across diverse tasks, including sorting, set operations, document merging, and open-domain question answering, we demonstrated that these FoT-augmented methods consistently outperform their baseline counterparts in terms of accuracy, semantic quality, and robustness under uncertainty. Notably, GoT⁺ exhibited the strongest overall performance, followed by ToT⁺, confirming the effectiveness of incorporating semantic fusion into multi-branch and graph-structured reasoning processes. Furthermore, our cost and token analyses showed that the improvements achieved by FoT come with minimal additional overhead, making it a practical and scalable solution. In summary, this work establishes FoT as a promising pathway toward interpretable, consistent, and extensible reasoning in LLMs, and paves the way for future research on integrating fusion-based reasoning with knowledge retrieval, multimodal inputs, and real-world decision-support applications.

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著作權聲明

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系所組別：資訊工程

研究生：蔡澍智

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